

MINXING PANG

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[GitHub](#) | [Google Scholar](#)

RESEARCH INTERESTS

AI for Healthcare · Biomedical Imaging · Spatial Omics · Computational Pathology · Deep Learning

My research develops deep learning methods that bridge routine histology imaging and molecular measurement, turning ubiquitous tissue images into rich, spatially resolved, multi-modal readouts for precision medicine.

EDUCATION

University of Pennsylvania, Philadelphia, PA, USA 2022 – 2026 (Expected)

Ph.D. in Applied Mathematics and Computational Science

Advisor: Kai Tan, Ph.D., Professor of Pediatrics

King Abdullah University of Science and Technology (KAUST), Saudi Arabia 2020 – 2022

M.S. in Applied Mathematics and Computational Science

Advisor: Jesper Tegnér, Ph.D., Professor of Bioscience

Nankai University, Tianjin, China 2016 – 2020

B.S. in Information and Computational Science

Advisor: Jishou Ruan, Ph.D., Professor of Mathematics

Thesis: *Modeling single-cell RNA-seq data using deep learning*

PUBLICATIONS

Name in **bold** indicates author. Google Scholar profile linked in header.

- [1] **Pang M**, Roy TK, Wu X, Tan K. (2025). CelloType: A Unified Model for Segmentation and Classification of Tissue Images. *Nature Methods*. [[Paper](#)] [[Codes](#)]
- [2] Bandyopadhyay S, Duffy M, Ahn K, Sussman J, **Pang M**, ..., and Tan K. (2024). Mapping the cellular biogeography of human bone marrow niches using single-cell transcriptomics and proteomic imaging. *Cell*. [[Paper](#)]
- [3] Yu W, Biyik-Sit R, Uzun Y, Chen CH, Thadi A, Sussman JH, **Pang M**, ..., and Tan K. (2025). Longitudinal single-cell multiomic atlas of high-risk neuroblastoma reveals chemotherapy-induced tumor microenvironment rewiring. *Nature Genetics*.

CONFERENCE PAPERS

- [4] **Pang M**, Tegnér J. (2020). Representation Learning and Translation between the Mouse and Human Brain using a Deep Transformer Architecture. *International Conference on Machine Learning (ICML) 2020 Workshop on Computational Biology*. **Contributed Talk** (Top 5%). [[PDF](#)] [[Codes](#)]

MANUSCRIPTS UNDER REVIEW / PREPRINTS

- [5] **Pang M**, Roy TK, Wu X, Tan K. (2025). A Query-driven generative AI synthesizes multi-modal spatial omics from histology. *Under review*. [[bioRxiv](#)]
- [6] **Pang M**, Su K, Li M. (2021). Leveraging information in spatial transcriptomics to predict super-resolution gene expression from histology images in tumors. [[bioRxiv](#)] [[Codes](#)]

INVITED TALKS & PRESENTATIONS

<i>Towards Unified Segmentation and Classification of Tissue Images</i>		
HTAN-HuBMAP Joint Meeting, Stanford University, CA		March 2024
Schmidt Center Symposium: Biomedical Science and AI, Koch Institute, MIT, MA		April 2025
<i>Generative AI synthesizes multi-modal spatial omics from histology</i>		
Shanghai Medical College, Fudan University, CN (Invited)		December 2024
School of Medicine, Zhejiang University, CN (Invited)		July 2025

HONORS & AWARDS

KAUST Fellowship	2020 – 2022
Outstanding Undergraduate Thesis of Nankai University (Top 3%)	2020
Nankai University Innovation Scholarship	2017, 2018
Provincial First Prize in Chinese Physics Competition	2015

MENTORING & TEACHING

Research Mentor , Tan Lab, Children’s Hospital of Philadelphia & UPenn	2022 – Present
Mentor junior graduate and undergraduate students on projects in computational pathology and spatial omics, guiding method development, coding, and manuscript preparation.	

TECHNICAL SKILLS

Advanced: Python, PyTorch, Deep Learning, Biomedical Imaging Analysis
Familiar: R, C++

REFERENCES

References available upon request.